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Installing and extending an ABAP MCP Server by vibe-coding with Claude Code and Google Antigravity

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k1000
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2026 Mar 06 5:40 PM

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ABAP Development

On one rainy morning I mentioned to my Google Antigravity Agent (Claude Opus 4.6 Thinking):

"I hear people are experimenting with connecting to SAP Systems via MCP. They use ABAP Development Tools API as a way in. Could you help me set up something like that for my ABAP Trial system?"

He searched the Web and found Mario Andreschak's MCP-ABAP-ADT solution on GitHub: <https://github.com/mario-andreschak/mcp-abap-adt>. Then he came up with the following implementation plan and asked a few questions:

Files Edited

Implementation Plan

Progress Updates

Expand all

1 Reading the GitHub README for mcp-abap-adt to understand exact setup steps and configuration

2 Writing the implementation plan for the user to review, then checking for Node.js and Git on the system

3 Writing the implementation plan artifact

Here's the setup plan — everything you need is already on your machine (Node.js, npm, Git are all present).

Before I proceed, I have two quick questions highlighted in the plan:

1. Your SAP Trial URL — is it the standard `http://vhcala4hci:8000` (Docker-based ABAP Platform Trial), or something different?

2. Which MCP client do you want to connect to?

Claude Desktop (recommended)

VS Code Cline extension

Also let me know if your credentials differ from the Trial defaults (`DEVELOPER` / `Download` , client `001`).

I approved the plan and answered the questions. After a while, he was done cloning Mario's Repository and installing:

All done! Here's what was set up:

✓ Cloned `mcp-abap-adt` → `~/Code/Working/mcp-abap-adt`

✓ Built the TypeScript project → `dist/index.js` ready

✓ Created `.env` with your Trial credentials (`:50000` , client `100` , user `DEVELOPER`)

✓ Registered the server in Claude Desktop's config

One thing to do now: → Quit and relaunch Claude Desktop

After restarting, open a new conversation and ask something like:

"Use the ABAP MCP tool to get the source code of program `RSDEMO00`"

You should see the  hammer icon indicating tool use and the MCP server name appear.

I updated the Password and Client in the `*.env` file and switched to Claude Code, who (Don't anthropomorphise computers, they hate it!) can act as an MCP Client. Claude Code was able to connect to my ABAP Trial using Mario's MCP Server.

> Used mcp-abap-adt integration

📄 The connection works! 🎉 But program `RSDEMO00` doesn't exist on this system. The available RSDEMO programs are:

Program	Description
<code>RSDEMO02</code>	Demo for Table Control
<code>RSDEMO_CALENDAR_CONTROL</code>	Calendar Control
<code>RSDEMO_CONTEXT_MENU_DYNPR</code>	Context Menu Dynpro
<code>RSDEMO_CUSTOM_CONTROL</code>	Custom Control
<code>RSDEMO_HTML_VIEWER</code>	HTML Viewer
<code>RSDEMO_PICTURE_CONTROL</code>	Picture Control
<code>RSDEMO_SPLITTER_CONTROL</code>	Splitter Control
... and more	

Would you like me to fetch the source of one of these instead? `RSDEMO02` is the closest match.

Please fetch that one.

> Used mcp-abap-adt integration

Here's the source code of `RSDEMO02` — a demo program for SAP Table Controls. ✓

It demonstrates:

- Displaying flight connection data (`SPFLI` table) in a table control
- Sorting by selected columns
- Inserting/deleting lines
- Modifying column properties (visibility, input/output, length, etc.)
- Global table control properties (fixed columns, grids, scroll, selection modes)

The program `RSDEMO00` doesn't exist on this A4H system — `RSDEMO02` was the closest match available.

Not having studied Mario's documentation, I never realised his was a read-only MCP Server. I asked Claude Code to add a comment line to an ABAP Program, just as a test. He said he couldn't do it on account of there being no WRITE, and offered to extend the MCP Server with WRITE capability. I said yes.

Please fetch the source code of the Program ZOO001.

> Used mcp-abap-adt integration

📄 The program `Z00001` exists but is essentially empty — just the report header with no logic:

```
REPORT Z00001.
```

Would you like me to do something with it? 🛠️

Please add a comment line to it. Just as a test.

📄 Unfortunately, the ABAP MCP server only has read-only tools available (GetProgram, GetClass, SearchObject, etc.). There's no tool to write or update program source code. 😞

To modify `Z00001`, you'd need to either:

1. Use **SAP GUI** (SE38/SE80) to edit it directly
2. Use **ADT in Eclipse/VS Code** (the Antigravity extension I see you have installed)
3. Add a write/update capability to the `mcp-abap-adt` MCP server

Would you like me to look into extending the MCP server with a write capability?

Yes, please.

It didn't take long for Claude to do this, but I hit my rate limit and had to wait until the afternoon (Claude Code sessions are 5-hour windows with limits on input/output tokens, which reset at the end of the window. In addition, there are weekly limits).

```

v Updated todo list
└─ Update Todos
   └─ Extend makeAdtRequest in utils.ts with extraHeaders parameter
   └─ Create handleWriteProgram.ts handler
   └─ Register WriteProgram tool in index.ts
   └─ Build and test

📄 All done! 🚀 The WriteProgram tool has been added. Here's what was changed:
1. src/lib/utils.ts - Added extraHeaders parameter to makeAdtRequest() and exported fetchCsrfToken()
2. src/handlers/handleWriteProgram.ts (new) - Implements the full ADT write flow: lock → write → unlock → activate
3. src/index.ts - Registered the WriteProgram tool with schema and routing

Next step: Restart Claude Desktop so the MCP server picks up the new build. After that, we can test by adding a comment to Z00001. 🔄



Please add a comment to the Program ZOO001.



> Used mcp-abap-adt integration

You've hit your limit · resets 1pm (Europe/🇩🇪)

```

There was a bit of a kerfuffle regarding the locking, but Claude Code managed to sort it all out by himself.

```

> Used mcp-abap-adt integration

🚀 WriteProgram MCP tool is working! Let me verify by reading the program back:

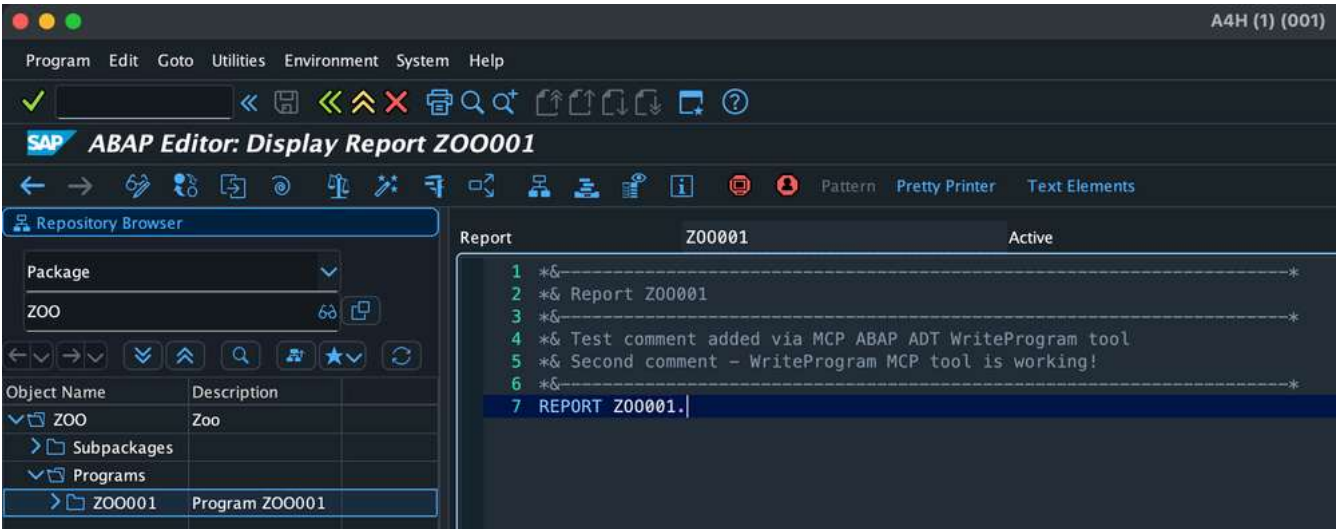
> Used mcp-abap-adt integration

✅ Everything is confirmed working! The WriteProgram MCP tool successfully:
1. Locked the program with a stateful session
2. Wrote the updated source code with the new comment
3. Unlocked the program
4. Activated it on the SAP system

The GetProgram verification confirms both comments are now in ZOO001's source. 🚀

```

And sure enough, the test ABAP Program was updated with a comment. Actually, two comments - one was there from before, when Claude was testing the API via CURL.



As I was again nearing my Claude Code usage limits, I switched back to Google Antigravity. I asked the agent there to add even more WRITE tools: For Classes and Interfaces, plus a tool to create Programs from scratch. My request was dispatched rather swiftly.

All done! Here's a summary of what was implemented:

4 New Write Tools Added to `mcp-abap-adt`

Tool	What it does
<code>TS WriteClass</code>	Update class source code (main, definitions, or implementations section)
<code>TS WriteInterface</code>	Update interface source code
<code>TS WriteInclude</code>	Update include source code
<code>TS CreateProgram</code>	Create a brand new ABAP program with source code + activation

These join the existing `TS WriteProgram` tool, giving you **5 write tools** plus the original 13 read tools = **18 tools total**.

Files Created/Modified

- **New:** `TS handleWriteClass.ts` , `TS handleWriteInterface.ts` , `TS handleWriteInclude.ts` , `TS handleCreateProgram.ts`
- **Modified:** `TS index.ts` (imports + tool registrations + dispatch)
- **Build:** Compiled with zero TypeScript errors

Next Step

Restart Claude Desktop so it picks up the rebuilt server. Then try something like:

"Create a new ABAP program called ZMCP_HELLO in package \$TMP that writes 'Hello from MCP!' to the screen"

Claude Desktop will use the `TS CreateProgram` tool to create it on your Trial system at .

I asked the Agent what other functionalities could he add to the MCP Server. As one would expect, he listed some of the many that are supported by the ADT API:

Great question! The SAP ADT REST API is quite extensive. Here's what else could be built, organized by category:

Object Creation (like `ts CreateProgram`)

Tool	ADT Endpoint	What it does
<code>CreateClass</code>	<code>POST /sap/bc/adt/oo/classes</code>	Create a new ABAP class
<code>CreateInterface</code>	<code>POST /sap/bc/adt/oo/interfaces</code>	Create a new interface
<code>CreateFunctionGroup</code>	<code>POST /sap/bc/adt/functions/groups</code>	Create a new function group
<code>CreateFunctionModule</code>	<code>POST /sap/bc/adt/functions/groups/{fg}/fmodules</code>	Create a function module inside a group
<code>CreatePackage</code>	<code>POST /sap/bc/adt/packages</code>	Create a development package
<code>CreateTable</code>	<code>POST /sap/bc/adt/ddic/dtl/sources</code>	Create a DDIC table/CDS view

More Write Tools

Tool	What it does
<code>WriteFunctionModule</code>	Update function module source
<code>WriteCDSView</code>	Update CDS view definition source
<code>WriteBehaviorDefinition</code>	Update RAP behavior definition

Lifecycle & Transport

Tool	ADT Endpoint	What it does
<code>ActivateObject</code>	<code>POST /sap/bc/adt/activation</code>	Activate any object by URI (standalone)
<code>CheckSyntax</code>	<code>POST /sap/bc/adt/checkruns</code>	Run syntax check before activation
<code>GetTransportRequests</code>	<code>GET /sap/bc/adt/cts/transportrequests</code>	List transport requests
<code>ReleaseTransport</code>	<code>POST /sap/bc/adt/cts/transportrequests/{id}</code>	Release a transport

Advanced Read / Analysis

Tool	What it does
<code>GetCDSView</code>	Read CDS view source
<code>GetBehaviorDefinition</code>	Read RAP behavior definition source
<code>GetServiceBinding</code>	Read OData service binding details
<code>GetWhereUsed</code>	Find all usages of an object (cross-references)
<code>GetObjectMetadata</code>	Get object properties (created by, last changed, package, etc.)
<code>RunAbapUnit</code>	Execute ABAP Unit tests and return results
<code>GetClassHierarchy</code>	Get inheritance tree of a class

Data & System

Tool	What it does
<code>ExecuteSQL</code>	Run an ABAP SQL query (via CDS or Open SQL console)
<code>GetSystemInfo</code>	Retrieve system name, version, kernel info
<code>GetUserInfo</code>	Get current user details and authorizations

All we have to do is ask 😊

PS:

- The SAP System was ABAP Cloud Developer Trial 2023,
- I used a Mac,
- Claude Code and Google Gemini paid plans. Not any fancy "Max" plans, just the regular 20-EUR-per-month ones.

LABELS

GenAI Assisted Content

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4 Comments



iabaper

Newcomer

2026 Apr 19 1:58 PM

...



0 Likes

SAP OP VERSION SUPPORTS THE MCP TOOLS.



patrick_weber11

Active Participant

2026 Apr 21 3:29 PM

...



0 Likes

Before others waste their tokens as well, do you provide the resulting MCP server somewhere?



k1000

Active Participant

2026 Apr 22 7:15 AM

...



1 Like

@patrick_weber11 Sorry, I don't have it anymore. I was just experimenting for the first time in my life, didn't save it.

But good news: There are several excellent, fully-built and actively developed community MCP ADT servers. The most famous one is this one: <https://github.com/oisee/vibing-steampunk>



Anil_Chauhan

Explorer

3 weeks ago

...



0 Likes

Hi @k1000 ,

This is really nice to use ADT service with LLM for Vibe coding, I am definitely looking forward to using this Repo.

Just wanted to check with you regarding SAP's API Policy FAQ, it clearly states to use ADT services by endorsed tool only.

How do you see this policy's impact on using ADT service for Vibe coding? If it is compliant with SAP's API Policy, it would help to build agents or with Vibe coding a lot can be solved with help of AI.

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